SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**

Question Count: 10

Code of Conduct

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: beffy

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

ANSWER

1. Explain the difference between HDFS and traditional file systems.

Answer:
HDFS (Hadoop Distributed File System) is designed to store very large datasets across multiple machines, whereas traditional file systems generally store data on a single server or storage system.

- **HDFS:** Distributed, highly fault-tolerant, scalable, and designed for large files.
- **Traditional file system:** Usually centralized and designed for local or network-based storage.
- HDFS uses **block storage and replication** to tolerate failures.

Interview point: HDFS is optimized for high-throughput access to big data, not low-latency access to small files.

2. Why is replication important in distributed storage systems?

Answer:
Replication creates multiple copies of data on different nodes. If one node fails, another copy can be used.

For example, with a replication factor of 3:

```
Data Block
├── Copy 1 → Node 1
├── Copy 2 → Node 2
└── Copy 3 → Node 3
```

This improves data availability, fault tolerance, and reliability. HDFS can also automatically create a new replica when a node fails.

3. How does MapReduce divide work between map and reduce phases?

Answer:
MapReduce divides a large processing task into two main phases:

- **Map phase:** Processes input data in parallel and produces intermediate key-value pairs.
- **Shuffle and Sort:** Groups intermediate data according to keys.
- **Reduce phase:** Processes each key and its associated values to generate the final result.

Example: For counting words, the mapper produces (word, 1), and the reducer adds all values for each word.

4. What role does YARN play in large-scale Hadoop processing?

Answer:
YARN (Yet Another Resource Negotiator) is Hadoop's resource management and job scheduling layer.

It:

- Allocates CPU and memory resources.

- Schedules applications.
- Manages containers.
- Allows multiple processing frameworks such as MapReduce and Spark to share the same cluster.

Architecture:

Applications

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YARN ResourceManager

↓

NodeManagers

↓

Containers

So, YARN makes the Hadoop cluster efficient and multi-user.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS	**SAN**
Network Attached Storage	Storage Area Network
Provides file-level access	Provides block-level access
Usually accessed through network file protocols	Usually accessed through specialized storage networks
Easier to manage	More complex
Suitable for shared files and documents	Suitable for databases and high-performance workloads

Simple explanation: NAS provides files, while SAN provides raw storage blocks to servers.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL stands for Extract, Transform, Load.

- **Extract:** Collect data from databases, files, applications, sensors, etc.
- **Transform:** Clean, validate, filter, and convert the data.
- **Load:** Store the processed data in a data warehouse, data lake, or analytics system.

Sources → Extract → Transform → Load → Analytics

ETL ensures that data is clean, consistent, and ready for analysis.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow and integration tool used to move and process data between different systems.

It provides:

- Data ingestion from multiple sources.
- Routing and filtering.
- Data transformation.
- Monitoring and flow management.
- Provenance tracking.
- Integration with databases, files, APIs, Hadoop and cloud systems.

Architecture example:

```
Database ─┐
       ├──→ NiFi → Transform → HDFS/Data Lake
Files ────┤
APIs ─────┘
```

NiFi is especially useful when an organization needs **controlled and monitored movement of data**.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:
Major failure points include:

1. **DataNode failure** → Use HDFS replication and automatic re-replication.
2. **NameNode failure** → Use NameNode HA with Active and Standby NameNodes.
3. **NodeManager failure** → YARN detects the failure and reschedules containers.
4. **Network failure** → Use redundant network paths and proper monitoring.
5. **Disk failure** → Use multiple disks and HDFS replication.
6. **Power/hardware failure** → Use redundant infrastructure and recovery procedures.

The main principle is to remove single points of failure through replication, HA and automatic recovery.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:
I would design the pipeline with security at every stage:

Data Sources

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Secure NiFi Ingestion

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Encryption / Authentication

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HDFS / Data Lake

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Access Control

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Processing & Analytics

Key measures:

- Use TLS/SSL for data in transit.
- Use strong authentication such as Kerberos where appropriate.
- Encrypt sensitive data at rest.
- Apply role-based access control.
- Maintain audit logs and data provenance.
- Validate and sanitize incoming data.
- Use secure backup and disaster recovery mechanisms.

This provides confidentiality, integrity and controlled access throughout the pipeline.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain it using a simple analogy:

"Think of Hadoop as a large team of workers. Instead of asking one person to process a huge amount of data, we divide the work among many workers and let them work at the same time."

For example:

Huge Data

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Divide into smaller tasks

↓

Many machines process simultaneously

↓

Combine the results

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Final Business Report

If the company gets more data, we can add more machines instead of replacing the entire system. HDFS keeps multiple copies of important data for reliability, while YARN manages available resources.